

Cancer is a large group of diseases characterized by uncontrolled cell growth and spread, caused by genetic mutations from environmental, lifestyle, or inherited factors. It is diagnosed via biopsies, imaging, and lab tests, with treatments including surgery, chemotherapy, radiation, targeted therapy, and immunotherapy. Latest research focuses on personalized medicine, improving immunotherapy, and early detection technologies.



Cleveland Clinic +7

Key Aspects of Cancer

- **Definition & Mechanism:** Cancer occurs when abnormal cells ignore signals to stop growing or die, forming tumors (malignant) that can invade nearby tissues and metastasize (spread) throughout the body.
- **Common Types:**
With over 200 types, common forms include breast, lung, colorectal, and prostate cancer.
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- **Causes & Risk Factors:** Around 90–95% of cases are due to environmental/lifestyle factors, including tobacco use (25–30%), diet and obesity (30–35%), infections (15–20%), and radiation.
- **Symptoms:** Signs are often vague, including unexpected weight loss, fatigue, fever, skin changes, or persistent pain.



Wikipedia +7

- **Diagnosis**
- **Biopsy:** Removing a small tissue sample to examine under a microscope is the most definitive method.
- **Imaging Tests:** X-rays, CT scans, MRIs, and PET scans help locate and monitor tumor growth.
- **Laboratory Tests:** Blood or urine tests can identify specific cells or substances indicating cancer.



National Cancer Institute (.gov) +4

- **Treatment Options**
- **Surgery:** Aims to remove the tumor locally.
- **Chemotherapy:** Systemic treatment using drugs to kill fast-growing cancer cells.
- **Radiotherapy:** Uses high-energy radiation to destroy cancer cells.
- **Targeted Therapy:** Targets specific changes or proteins within cancer cells.
- **Immunotherapy:** Boosts the immune system to fight cancer.



Healthline +4

- **Latest Advancements & Research**
- **Personalized/Precision Medicine:** Tailoring treatments based on the specific genetic profile of a tumor.
- **Improved Immunotherapy:** Developing CAR-T cell therapy and checkpoint inhibitors to make the immune system more effective.
- **Early Detection:** Advancements in liquid biopsies (testing for DNA in blood) to catch cancers earlier.



World Health Organization (WHO) +4

- Preventive measures, including quitting smoking, healthy diet, vaccination (e.g., HPV, HBV), and limiting UV exposure, can prevent 30% to 50% of cancer deaths